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Research article

Decentralized triangular relative localization of multiple UAVs based on relative range and inertial measurements☆

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ABSTRACT

In this paper, a novel algorithm for cooperative relative localization of multiple Unmanned Aerial Vehicles (UAVs) is proposed based on relative range and inertial measurements. In this algorithm, a relative motion estimation model is established for each group of three UAVs that can form a triangle in space. Each group member estimates the relative position and heading angle of the other group members and shares the estimation results with the other group members. When members satisfy the triangle law among their estimated relative position vectors, they can estimate more accurately. After analyzing the observability of the presented model, the necessary conditions for observability are also given, which are less restrictive compared to previous methods. Then, the presented method is compared with the decentralized consensus-based Kalman filter approach. The results of both methods are analyzed in the presence of noise and disturbances, and under the condition of disconnection between the members.

1. Introduction

Cooperation of UAVs provides advantages that cannot be achieved by a single UAV. A cooperative swarm of UAVs has the potential to execute missions more efficiently and quickly than a single UAV. It can also increase mission reliability and scalability and decrease mission cost. Multi-UAVs have many applications, such as surveillance [1,2], object localization [3], formation flight [4–7], etc. To perform a reliable and versatile swarm mission, it is usually necessary for the UAVs to sense each other and interact locally. Some of the cooperative flight missions (e.g. close formation flights) require accurate measurement or estimation of the relative position between the agents. Therefore, there is a great motivation to develop relative position estimation methods that take sensor limitations, reliability, and computation capabilities into account.

There are two major categories of existing localization systems: absolute and relative. Global Navigation Satellite Systems (GNSSs) such as GPS [4,8] and motion capture systems [9] are the most well-known absolute localization systems. However, in GPS-denied environments, GPS may not work and its accuracy may not be sufficient for operations where UAVs fly close to each other [10]. Moreover, a motion capture system can provide accurate positioning information, but it is typically limited to laboratory environments and can be costly to implement. On the other hand, relative localization refers to fusing measurements from different sensors (such as camera, Inertial Measurement Unit (IMU), lidar, and Ultra-WideBand (UWB) ranging sensors) mounted on UAVs to determine relative locations between each pair of UAVs. For instance, a laser-camera system has been utilized to address the multi-robot map-alignment problem in Simultaneous Localization and Mapping (SLAM) systems [11]. Visual-Inertial-UWB based estimation system has been used to estimate the relative position of Multi-UAVs [12]. Inertial Navigation System (INS), odometer, and UWB have been utilized for positioning in GPS-denied coal mine environments [13].

Vision-based methods, such as onboard camera-based or infrared sensor systems, have a limited field of view for cooperative UAVs. UAVs with an array of sensors or active tracking of neighboring agents can address the limited field of view issue. However, neither solution is ideal for UAVs as they are heavy and rely on environments for accuracy. Hence, a wireless radio transceiver such as UWB, an inherently omni-directional sensor, has been proposed as a more suitable alternative for relative positioning in cooperative UAVs [14]. Ultra-wideband radio technology has made it more accessible to measure the distance (with an advertised error of less than 10 cm) between UAVs due to its cost-effectiveness, power-efficiency, and ability to function in GPS-denied environments. However, a single distance measurement cannot provide a full three-dimensional relative positioning estimate, requiring

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additional information to achieve an observable relative positioning solution [15].

Range-based positioning is commonly achieved by measuring the relative range of landmarks with known positions (named as anchors) [16], but estimating the relative position between moving agents without external reference is more challenging. Additional sensors, such as cameras or multiple UWB tags, have been used to achieve relative positioning from distance measurements. However, vision-based methods require substantial computational power [12]. Using multiple tags has been proposed as a solution to the unobservability problem. Several papers have demonstrated the use of multiple tags to achieve observable relative positioning, including the use of two tags and a 9-axis IMU [17], four tags on a quadrotor for tracking a person [18], and three tags on a hexacopter to localize a quadrotor [19]. For multiple UWB tags, the optimal location of each UAV to enhance estimation performance has been investigated in [20]. Also, the observability conditions have been derived in [15] with the help of graph theory. According to [21,22], the observability of the relative position can be achieved with single-distance measurements and velocity information, but only when the agents are in continuous relative motion and satisfy a persistency of excitation condition.

There are two main approaches to estimating relative position based on relative range: optimization-based and filter-based. Optimization-based methods determine possible sensor positions based on distance geometry, but noisy measurements and sensor motion can limit their accuracy [23]. The sliding window concept [24], efficient cost functions [25], and pre-filtering [26] have been proposed as solutions to overcome these limitations. In order to enhance the computational efficiency of optimization-based methods, cost functions are reformulated as semidefinite programming problems [27,28] and convex cost functions are defined [29]. However, nonlinear optimization methods require large computational resources due to their centralized architectures, making them unsuitable for embedded applications with limited resources.

Filtering-based approaches, such as the extended Kalman filter (EKF), use inertial sensors to predict robot motion and then update beliefs with measurements [15,30]. Recursive filtering algorithms like EKF are more suitable for decentralized architectures [17]. Reliability and ability to distribute computational load among members have made decentralized methods more popular in recent years. Besides, it reduces the complexity of the algorithm for each agent. This means that the success of a mission is no longer dependent on a single leader. Furthermore, there is an increasing demand for in-network data processing tools and algorithms, providing high-performance online estimation while reducing the communication load among sensor nodes, being robust to sensor node failures or replacements, and suitable for distributed control applications [31]. These requirements highlight the need for efficient and robust decentralized estimation methods. Consensus-based EKFs are decentralized methods that use consensus techniques to enable distributed estimation through two consecutive steps. First, the EKF is applied to estimate the relative positions of the agents. Second, a correction step is performed to enforce the geometric constraints to be satisfied [32–34]. Consensus-based EKFs can be used in large-scale networks with limited communication and computation resources. However, these methods are susceptible to bias or convergence to the wrong solution due to the presence of faulty or malicious agents. Furthermore, consensus-based EKFs require parameter tuning and can be slow to converge.

In some previous works, in addition to one IMU, multiple UWBs have been used on each UAV to provide system observability and improve the convergence rate and accuracy of relative position estimation between UAVs [15,24,30]. However, the current study proposes a fast decentralized method that uses a single UWB and IMU measurements, on each UAV, to estimate accurately relative position between UAVs in a swarm. In summary, the main contributions include:

- A relative motion model is developed for the problem of estimating the relative position vectors of a group of three UAVs so that the estimated information of other members can also be considered in the measurement vector of each UAV.
- Unlike decentralized consensus-based EKFs, where the estimation operation is typically performed in two steps (first estimation and then correction based on the geometric constraints), the proposed method performs estimation while simultaneously ensuring satisfaction of the geometric constraints.
- Unlike previous researches, fusion of individual measurements with the information estimated by the others, makes the system observable, even during the hovering and parallel flights.

The rest of this paper is structured as follows. Section 2 outlines the problem statement and relative model. The observability analysis is presented in Section 3, while Section 4 discusses the decentralized estimation strategy based on the EKF. Simulations are carried out in Section 5. Finally, the conclusion of the paper is given in Section 6.

2. Problem statement and relative model

This section introduces the problem statement and outlines the assumptions made in the study. The developed dynamics model, which consists of the dynamics of three UAVs, is also presented.

2.1. Problem definition

In this article, the problem of estimating the relative localization of multiple UAVs in a GPS-denied environment is considered. The problem of estimating relative position and orientation is simplified to only motion in the horizontal plane. In this case, using UWB sensors mounted on each UAV, the distance between UAVs can be measured. Moreover, the UWB and the IMU measurements of each UAV are shared among the UAVs.

2.2. Relative motion model

In this paper, the inertial frame is denoted by $\mathcal{I}$. The body-fixed frame, denoted by β_i, is assumed to originate at the center of gravity of the ith quadrotor, where the UWB and IMU sensors are installed. The axes in the body frame point to the head, right, and down. The z-axis of each quadrotor's frame is also assumed to be parallel to the z-axis of the inertial frame, as quadrotors fly in a near-hover state with small roll (ϕ) and pitch (θ) angles. Rotation angle along the z-axis of the ith quadrotor, denoted by ψ_i, is defined as the heading angle. The rate of changing heading angle is denoted by ω and can be calculated as follows:

$$\tilde{\omega}_i = \frac{\tilde{q}_i \sin(\tilde{\phi}_i) + \tilde{r}_i \cos(\tilde{\phi}_i)}{\cos(\tilde{\theta}_i)} \tag{1}$$

where $\tilde{q}$ and $\tilde{r}$ are the pitch and yaw rates of the quadrotor measured by a rate gyroscope, IMU. As shown in 1, The 2D vector $\mathbf{P}_{ij}^{\beta_i}$ denotes the relative position of the ith quadrotor with respect to the jth quadrotor expressed in frame β_i. The $\Delta\psi_{ij} = \psi_j - \psi_i$ represents the relative heading angle between quadrotors i and j. The 2D vectors $\mathbf{v}_i^{\beta_i}$ and $\tilde{\mathbf{a}}_i^{\beta_i}$ also represent the velocity and specific force measured by mounted IMU of the ith quadrotor expressed in frame β_i, respectively. In the following, by defining the state vector, $\mathbf{X}_{i\lhd_k^j} \in \mathbb{R}^{12}$, as

$$\mathbf{X}_{i\lhd_k^j} = \left[[\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}}, [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}}, \Delta\psi_{ij}, \Delta\psi_{ik}, [\mathbf{v}_i^{\beta_i}]^{\mathrm{T}}, [\mathbf{v}_j^{\beta_j}]^{\mathrm{T}}, [\mathbf{v}_k^{\beta_k}]^{\mathrm{T}}\right]^{\mathrm{T}} \tag{2}$$

and input vector $\mathbf{U}_{i\lhd_k^j} \in \mathbb{R}^9$ as

$$\mathbf{U}_{i\lhd_k^j} = \left[\tilde{\omega}_i, \tilde{\omega}_j, \tilde{\omega}_k, [\tilde{\mathbf{a}}_i^{\beta_i}]^{\mathrm{T}}, [\tilde{\mathbf{a}}_j^{\beta_j}]^{\mathrm{T}}, [\tilde{\mathbf{a}}_k^{\beta_k}]^{\mathrm{T}}\right]^{\mathrm{T}} \tag{3}$$

The relative motion dynamics for the ith quadrotor, which attempts to estimate the relative position and heading angle with respect to the

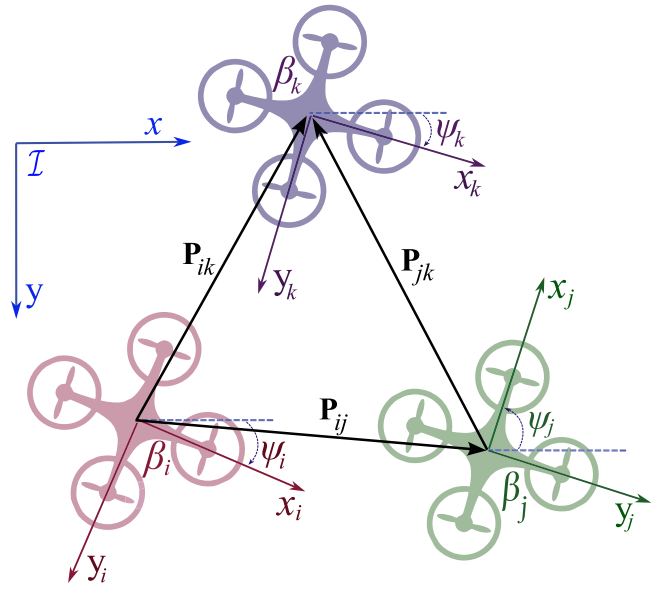


Fig. 1. A geometric representation of the relative positions and heading angles of three quadrotors. It should be noted that, in this case the sign of ψ_j is negative.

jth and kth quadrotors, are developed as follows based on the model proposed in [22] and [32].

$$\dot{\mathbf{X}}_{i\lhd_k^j} = \mathbf{f}\left(\mathbf{X}_{i\lhd_k^j}, \mathbf{U}_{i\lhd_k^j}\right) = \begin{bmatrix} \mathbf{R}^{\beta_i\beta_j}\mathbf{v}_j^{\beta_j} - \tilde{\boldsymbol{\Omega}}_i\mathbf{P}_{ij}^{\beta_i} - \mathbf{v}_i^{\beta_i} \\ \mathbf{R}^{\beta_i\beta_k}\mathbf{v}_k^{\beta_k} - \tilde{\boldsymbol{\Omega}}_i\mathbf{P}_{ik}^{\beta_i} - \mathbf{v}_i^{\beta_i} \\ \tilde{\omega}_j - \tilde{\omega}_i \\ \tilde{\omega}_k - \tilde{\omega}_i \\ \tilde{\mathbf{a}}_i^{\beta_i} - \boldsymbol{\Omega}_i\mathbf{v}_i^{\beta_i} \\ \tilde{\mathbf{a}}_j^{\beta_j} - \boldsymbol{\Omega}_j\mathbf{v}_j^{\beta_j} \\ \tilde{\mathbf{a}}_k^{\beta_k} - \boldsymbol{\Omega}_k\mathbf{v}_k^{\beta_k} \end{bmatrix} \tag{4}$$

where $\tilde{\boldsymbol{\Omega}}_i$ is the 2D skew-symmetric matrix of $\tilde{\omega}_i$ and is equal to:

$$\boldsymbol{\Omega}_i = \begin{bmatrix} 0 & -\tilde{\omega}_i \\ \tilde{\omega}_i & 0 \end{bmatrix} \tag{5}$$

The $\mathbf{R}^{\beta_i\beta_j}$ is the 2D rotation matrix from β_j to β_i:

$$\mathbf{R}^{\beta_i\beta_j} = \begin{bmatrix} \cos(\Delta\psi_{ij}) & -\sin(\Delta\psi_{ij}) \\ \sin(\Delta\psi_{ij}) & \cos(\Delta\psi_{ij}) \end{bmatrix} \tag{6}$$

Finally, the measurement model, $\mathbf{y}_{i\lhd_k^j} \in \mathbb{R}^{12}$, for the estimation is defined as follows:

$$\begin{aligned} \mathbf{y}_{i\lhd_k^j} &= \mathbf{h}\left(\mathbf{X}_{i\lhd_k^j}\right) \\ &= \left[\left\|\mathbf{P}_{ij}^{\beta_i}\right\|, \left\|\mathbf{P}_{ik}^{\beta_i}\right\|, \mathbf{P}_{ik}^{\beta_i} - \mathbf{P}_{ij}^{\beta_i}, \Delta\psi_{ij}, \Delta\psi_{ik}, \mathbf{v}_i^{\beta_i}, \mathbf{v}_j^{\beta_j}, \mathbf{v}_k^{\beta_k}\right]^{\mathrm{T}} \end{aligned} \tag{7}$$

where $\|.\|$ is the Euclidean norm. Based on the information shared by the other two members of the group, $\mathbf{P}_{ik}^{\beta_i} - \mathbf{P}_{ij}^{\beta_i}$ can be determined. Indeed, as depicted in Fig. 1, $\mathbf{P}_{ik} - \mathbf{P}_{ij} = \mathbf{P}_{jk}$, where $\mathbf{P}_{jk}$ is obtained through the estimation process, performed by the jth and kth UAVs. In Section 4.2, more details are provided.

3. Observability analysis

Observability analysisocal weak observability is a property of a dynamical system that determines whether a particular state can be distinguished from others in its neighborhood based on evaluating measurements of the system [35]. This property can be investigated by checking the rank of the observability matrix defined using lie derivatives. First, in this section, the observability condition based on Lie derivatives is briefly explained. Then, the observability conditions for the proposed model are determined.

3.1. Observability based on Lie derivatives

A nonlinear dynamic system can be described as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \tag{8}$$

$$\mathbf{y} = \mathbf{h}(\mathbf{x}) \tag{9}$$

where $\mathbf{x} = [x_1, x_2, \ldots, x_n]^{\mathrm{T}} \in \mathbb{R}^n, \mathbf{u} = [u_1, u_2, \ldots, u_n]^{\mathrm{T}} \in \mathbb{R}^l$, and $\mathbf{y} = [y_1, y_2, \ldots, y_m]^{\mathrm{T}} \in \mathbb{R}^m$ are the vectors of states, control inputs and measurements, respectively. Also, $\mathbf{f}$ and $\mathbf{h}$ are nonlinear functions. The observability matrix, $\mathcal{O} \in \mathbb{R}^{mn}\times\mathbb{R}^n$, based on nth-order of Lie derivatives [35] can be written as

$$\mathcal{O}(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}}\left[\mathcal{L}_f^0\mathbf{h}(\mathbf{x}), \mathcal{L}_f^1\mathbf{h}(\mathbf{x}), \ldots, \mathcal{L}_f^{n-1}\mathbf{h}(\mathbf{x})\right]^{\mathrm{T}} \tag{10}$$

where Lie derivatives, $\mathcal{L}_f^i\mathbf{h}(\mathbf{x}) \in \mathbb{R}^{m\times 1}$, can be calculated as follows:

$$\mathcal{L}_f^i\mathbf{h}(\mathbf{x}) = \frac{\mathcal{L}_f^{i-1}\mathbf{h}(\mathbf{x})}{\partial \mathbf{x}}\mathbf{f}(\mathbf{x}, \mathbf{u}) \text{ with } \mathcal{L}_f^0\mathbf{h}(\mathbf{x}) = \mathbf{h}(\mathbf{x}) \tag{11}$$

In order to investigate the local weak observability of nonlinear systems, high-order Lie derivatives should be added to the observability matrix until the matrix becomes full rank, $\mathrm{rank}(\mathcal{O}) = n$. This system will be unobservable if its rank is not complete, despite the existence of all high-order Lie derivatives.

3.2. Observability analysis of relative motion model

As stated in the previous section, for the dynamic model presented in (4) with 12 states, the rank of the observability matrix needs to be equal to 12 so that the system becomes observable. In this section, the necessary conditions for observability are determined using at least a second-order Lie derivative. For ease of derivation, the measurement vector in (7) is modified as follows. This modification replaces the $\|\mathbf{P}\|$ with the $\frac{1}{2}\mathbf{P}^{\mathrm{T}}\mathbf{P}$, which contains the same information, so it does not affect the observability [36].

$$\begin{aligned} &\mathbf{h}\left(\mathbf{X}_{i\lhd_k^j}\right) \\ &= \left[\tfrac{1}{2}[\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}}\mathbf{P}_{ij}^{\beta_i}, \tfrac{1}{2}[\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}}\mathbf{P}_{ik}^{\beta_i}, \mathbf{P}_{ik}^{\beta_i} - \mathbf{P}_{ij}^{\beta_i}, \Delta\psi_{ij}, \Delta\psi_{ik}, \mathbf{v}_i^{\beta_i}, \mathbf{v}_j^{\beta_j}, \mathbf{v}_k^{\beta_k}\right]^{\mathrm{T}} \end{aligned} \tag{12}$$

The first entry in the observability matrix is equal to:

$$\frac{\partial \mathcal{L}_f^0\mathbf{h}(\mathbf{X}_{i\lhd_k^j})}{\partial \mathbf{X}_{i\lhd_k^j}} = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times 2} & 0 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{1\times 2} & [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & 0 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ -\mathbf{I}_{2\times 2} & \mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & 1 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & 0 & 1 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{I}_{2\times 2} \end{bmatrix} \tag{13}$$

where $\mathbf{I}$ and $\mathbf{0}$ represent an identity and null matrix, respectively. Considering $\frac{\partial \mathbf{P}^{\mathrm{T}}\boldsymbol{\Omega}\mathbf{P}}{\partial \mathbf{P}} = \mathbf{0}_{1\times 2}$ since $\boldsymbol{\Omega}$ is a skew-symmetric matrix, the

second entry in the observation matrix can be calculated as follows:

$$\frac{\partial L_f^1 \mathbf{h}(\mathbf{X}_{i\lhd_k^j})}{\partial \mathbf{X}_{i\lhd_k^j}} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0}_{1\times2} & B_1 & 0 & [& \mathbf{C}_1 &] \\ \mathbf{0}_{1\times2} & \mathbf{A}_2 & 0 & B_2 & [& \mathbf{C}_2 &] \\ \tilde{\mathbf{\Omega}}_i & -\tilde{\mathbf{\Omega}}_i & \mathbf{D}_1 & \mathbf{D}_2 & \mathbf{0}_{2\times2} & -\mathbf{R}^{\beta_i\beta_j} & \mathbf{R}^{\beta_i\beta_k} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & -\tilde{\mathbf{\Omega}}_i & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times2} & -\tilde{\mathbf{\Omega}}_j & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & -\tilde{\mathbf{\Omega}}_k \end{bmatrix} \tag{14}$$

where $\mathbf{A}_i, B_i, \mathbf{C}_i$, and $\mathbf{D}_i$ are defined as follows:

$$\mathbf{A}_1 = [\mathbf{v}_j^{\beta_j}]^{\mathrm{T}}[\mathbf{R}^{\beta_i\beta_j}]^{\mathrm{T}} - [\mathbf{v}_i^{\beta_i}]^{\mathrm{T}} \tag{15}$$

$$\mathbf{A}_2 = [\mathbf{v}_k^{\beta_k}]^{\mathrm{T}}[\mathbf{R}^{\beta_i\beta_k}]^{\mathrm{T}} - [\mathbf{v}_i^{\beta_i}]^{\mathrm{T}} \tag{16}$$

$$B_1 = [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}}\mathbf{\Gamma}\mathbf{R}^{\beta_i\beta_j}\mathbf{v}_j^{\beta_j} \tag{17}$$

$$B_2 = [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}}\mathbf{\Gamma}\mathbf{R}^{\beta_i\beta_k}\mathbf{v}_k^{\beta_k} \tag{18}$$

$$\mathbf{C}_1 = \begin{bmatrix} -[\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}}\mathbf{R}^{\beta_i\beta_j} & \mathbf{0}_{1\times2} \end{bmatrix} \tag{19}$$

$$\mathbf{C}_2 = \begin{bmatrix} -[\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}}\mathbf{R}^{\beta_i\beta_k} \end{bmatrix} \tag{20}$$

$$\mathbf{D}_1 = -\mathbf{\Gamma}\mathbf{R}^{\beta_i\beta_j}\mathbf{v}_j^{\beta_j} \tag{21}$$

$$\mathbf{D}_2 = \mathbf{\Gamma}\mathbf{R}^{\beta_i\beta_k}\mathbf{v}_k^{\beta_k} \tag{22}$$

where $\mathbf{\Gamma}$ is defined as follows:

$$\mathbf{\Gamma} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{23}$$

The observability matrix can be constructed as follows by combining (13) and the first two rows of (14) (since these are the only rows that can contribute to increasing the rank of the observability matrix whereas the rest are linearly dependent on some rows of (14), namely the rows of two vectors $\mathbf{\Omega}$ and $\mathbf{I}_{2\times2}$).

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{1\times2} & [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & 0 & 0 & \mathbf{0}_{1\times6} \\ -\mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times6} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 1 & 0 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 0 & 1 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{6\times2} & \mathbf{0}_{6\times2} & \mathbf{0}_{6\times1} & \mathbf{0}_{6\times1} & \mathbf{I}_{6\times6} \\ \mathbf{A}_1 & \mathbf{0}_{1\times2} & B_1 & 0 & \mathbf{C}_1 \\ \mathbf{0}_{1\times2} & \mathbf{A}_2 & 0 & B_2 & \mathbf{C}_2 \end{bmatrix} \tag{24}$$

To simplify calculating the rank of $\mathcal{O}(\mathbf{x})$, the elementary row operations include $-B_1R_5 + R_{13} \to R_{13}$ and $-B_2R_5 + R_{14} \to R_{14}$ are performed as follows. R_n represents nth row of the matrix $\mathcal{O}$.

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{1\times2} & [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & 0 & 0 & \mathbf{0}_{1\times2} \\ -\mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times6} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 1 & 0 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 0 & 1 & \mathbf{0}_{1\times6} \\ \mathbf{0}_{6\times2} & \mathbf{0}_{6\times2} & \mathbf{0}_{6\times1} & \mathbf{0}_{6\times1} & \mathbf{I}_{6\times6} \\ \mathbf{A}_1 & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{C}_1 \\ \mathbf{0}_{1\times2} & \mathbf{A}_2 & 0 & 0 & \mathbf{C}_2 \end{bmatrix} \tag{25}$$

By applying similar elementary row operations to the last rows, we can transform $\mathbf{C}_1$ and $\mathbf{C}_2$ into zero matrices. Additionally, by interchanging some rows, the decomposed form of $\mathcal{O}$ can be written as follows:

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ \mathbf{0}_{1\times2} & [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times8} \\ -\mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times8} \\ \mathbf{A}_1 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ \mathbf{0}_{1\times2} & \mathbf{A}_2 & \mathbf{0}_{1\times8} \\ \mathbf{0}_{8\times2} & \mathbf{0}_{8\times2} & \mathbf{I}_{8\times8} \end{bmatrix} \tag{26}$$

Suppose $[\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} = [Px_{ij}, Py_{ij}]^{\mathrm{T}}, [\mathbf{P}_{ik}^{\beta_k}]^{\mathrm{T}} = [Px_{ik}, Py_{ik}]^{\mathrm{T}}, \mathbf{A}_1 = [Ax_1, Ay_1]^{\mathrm{T}}$, and $\mathbf{A}_2 = [Ax_2, Ay_2]^{\mathrm{T}}$. Then, matrix $\mathcal{O}$ can be written as:

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} Px_{ij} & Py_{ij} & 0 & 0 & \mathbf{0}_{1\times8} \\ 0 & 0 & Px_{ik} & Py_{ik} & \mathbf{0}_{1\times8} \\ -1 & 0 & 1 & 0 & \mathbf{0}_{1\times8} \\ 0 & -1 & 0 & 1 & \mathbf{0}_{1\times8} \\ Ax_1 & Ay_1 & 0 & 0 & \mathbf{0}_{1\times8} \\ 0 & 0 & Ax_2 & Ay_2 & \mathbf{0}_{1\times8} \\ \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{I}_{8\times8} \end{bmatrix} \tag{27}$$

Now the two steps of elementary rows operations can be applied. Step 1: $-Px_{ik}R_3 + R_2 \to R_2$ and $-Ax_2R_3 + R_6 \to R_6$

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} Px_{ij} & Py_{ij} & 0 & 0 & \mathbf{0}_{1\times8} \\ Px_{ik} & 0 & 0 & Py_{ik} & \mathbf{0}_{1\times8} \\ -1 & 0 & 1 & 0 & \mathbf{0}_{1\times8} \\ 0 & -1 & 0 & 1 & \mathbf{0}_{1\times8} \\ Ax_1 & Ay_1 & 0 & 0 & \mathbf{0}_{1\times8} \\ Ax_2 & 0 & 0 & Ay_2 & \mathbf{0}_{1\times8} \\ \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{I}_{8\times8} \end{bmatrix} \tag{28}$$

Step 2: $-Py_{ik}R_4 + R_2 \to R_2$ and $-Ay_2R_4 + R_6 \to R_6$

$$\mathcal{O}(\mathbf{x}) = \begin{bmatrix} Px_{ij} & Py_{ij} & 0 & 0 & \mathbf{0}_{1\times8} \\ Px_{ik} & Py_{ik} & 0 & 0 & \mathbf{0}_{1\times8} \\ -1 & 0 & 1 & 0 & \mathbf{0}_{1\times8} \\ 0 & -1 & 0 & 1 & \mathbf{0}_{1\times8} \\ Ax_1 & Ay_1 & 0 & 0 & \mathbf{0}_{1\times8} \\ Ax_2 & Ay_2 & 0 & 0 & \mathbf{0}_{1\times8} \\ \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{0}_{8\times1} & \mathbf{I}_{8\times8} \end{bmatrix} = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ -\mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times8} \\ \mathbf{A}_1 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ \mathbf{A}_2 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times8} \\ \mathbf{0}_{8\times2} & \mathbf{0}_{8\times2} & \mathbf{I}_{8\times8} \end{bmatrix} \tag{29}$$

The 10 rows of the simplified form of the $\mathcal{O}$ are constructed with the identity matrix, which shows that the minimum rank of the matrix is 10. It is enough to find two more independent linear rows from the rest of the other 4 rows so that the rank of the matrix reaches 12, and satisfies the local observability criteria. As a result, the matrix below must contain two independent rows.

$$C = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} \\ [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} \\ \mathbf{A}_1 \\ \mathbf{A}_2 \end{bmatrix} = \begin{bmatrix} [\mathbf{P}_{ij}^{\beta_i}]^{\mathrm{T}} \\ [\mathbf{P}_{ik}^{\beta_i}]^{\mathrm{T}} \\ [\mathbf{v}_j^{\beta_j}]^{\mathrm{T}}[\mathbf{R}^{\beta_i\beta_j}]^{\mathrm{T}} - [\mathbf{v}_i^{\beta_i}]^{\mathrm{T}} \\ [\mathbf{v}_k^{\beta_k}]^{\mathrm{T}}[\mathbf{R}^{\beta_i\beta_k}]^{\mathrm{T}} - [\mathbf{v}_i^{\beta_i}]^{\mathrm{T}} \end{bmatrix} \tag{30}$$

In order to provide more practical insight, three sets of observability conditions can be derived, each of which guarantees at least two independent rows in matrix C. These conditions are as follows:

- First set

$$\mathbf{P}_{ij} \neq \mathbf{0}_{2\times 1} \tag{31a}$$

$$\mathbf{v}_i \neq \mathbf{0}_{2\times 1} \text{ or } \mathbf{v}_j \neq \mathbf{0}_{2\times 1} \tag{31b}$$

$$\mathbf{v}_i^{\beta_i} \neq \mathbf{R}^{\beta_i \beta_j} \mathbf{v}_j^{\beta_j} \tag{31c}$$

- Second set

$$\mathbf{P}_{ik} \neq \mathbf{0}_{2\times 1} \tag{32a}$$

$$\mathbf{v}_i \neq \mathbf{0}_{2\times 1} \text{ or } \mathbf{v}_k \neq \mathbf{0}_{2\times 1} \tag{32b}$$

$$\mathbf{v}_i^{\beta_i} \neq \mathbf{R}^{\beta_i \beta_k} \mathbf{v}_k^{\beta_k} \tag{32c}$$

- Third set

$$\mathbf{P}_{ij}^{\beta_i} \neq \nu \mathbf{P}_{ik}^{\beta_i} \tag{33}$$

where ν is an arbitrary constant.

Only one set of conditions must be satisfied to make the system weakly observable. According to a recent study in [22], in a group of three quadrotors, the first and second set of conditions should be satisfied, however, we found that satisfying only one is sufficient in our model. In addition, according to the third set of conditions, if the three quadrotors are not collinear (i.e., not positioned in a straight line), the system will be observable even when the quadrotors are hovering or moving in parallel. This indicates that the model presented in this paper, which incorporates estimated information of UAVs in addition to their measurements, will remain observable under a wide range of conditions.

In the proposed relative motion model, Eq. (4), differential equations $\Delta\dot{\psi}_{ij}$ and $\Delta\dot{\psi}_{ik}$ are independent of relative positions. Thus, if only relative heading angles are considered in the relative dynamics, the proposed relative motion model and its measurement model can be simplified as

$$\dot{\mathcal{X}}_{i\lhd_k^j} = \mathcal{F}\left(\mathcal{X}_{i\lhd_k^j}, \mathbf{U}_{i\lhd_k^j}\right) = \begin{bmatrix} \tilde{\omega}_j - \tilde{\omega}_i \\ \tilde{\omega}_k - \tilde{\omega}_i \end{bmatrix} \tag{34}$$

$$\mathcal{Y}_{i\lhd_k^j} = \mathcal{H}\left(\mathcal{X}_{i\lhd_k^j}\right) = \begin{bmatrix} \Delta\psi_{ij} \\ \Delta\psi_{ik} \end{bmatrix} \tag{35}$$

where the state vector, $\mathcal{X}_{i\lhd_k^j}$, is defined as

$$\mathcal{X}_{i\lhd_k^j} = \left[\Delta\psi_{ij}, \Delta\psi_{ik}\right]^{\mathrm{T}} \tag{36}$$

In order to investigate observability of this simplified 2nd order model, the first entry in the observability matrix based on Lie derivatives can be expressed as

$$\frac{\partial \mathcal{L}_{\mathcal{F}}^0 \mathcal{H}(\mathcal{X}_{i\lhd_k^j})}{\partial \mathcal{X}_{i\lhd_k^j}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{37}$$

Therefore, the observability matrix of the simplified model is full rank, and the relative heading angles are observable in any situation.

4. Decentralized estimation strategy based on EKF

When enough information on distance, accelerometer, and rates measured by sensors is not provided for the estimator (e.g., the observability conditions may be violated), it may diverge or even converge to a geometrically incorrect answer (e.g., the estimated vectors of the relative position of the quadrotors do not obey the triangle law). In this section, the proposed cooperative estimation method based on EKF is explained in detail. First, the extended Kalman filter is briefly explained, and then the configuration of quadrotors interaction is presented to enhance estimation accuracy through geometrical topology.

4.1. Extended Kalman filter

The Extended Kalman Filter (EKF) is an extension of the Kalman filter to handle nonlinear dynamic systems. EKF exhibits local convergence for a wide range of nonlinear systems. Moreover, it has been shown in [37] that if the initial estimation error of the filter is set within a reasonable range, the error diminishes exponentially over time, eventually reaching zero. In this method, the standard Kalman filter is used to solve the estimation problem using a linearized model of nonlinear dynamics of a system around the current estimate of the states. Following are the two steps of the EKF for propagating and updating estimation and covariance:

- **Propagation step**: Based on the previously estimated state and covariance matrix, a linearized model of a nonlinear dynamic system is used to propagate the estimate of the state and the covariance matrix forward in time. The prior predicted state estimate and covariance matrix at instant t_k are calculated using discretized equations as follows:
Priori state prediction:

$$\hat{\mathbf{X}}^-_{i\lhd_k^j,t_k} = \mathbf{F}_{i\lhd_k^j,t_k} \hat{\mathbf{X}}_{i\lhd_k^j,t_{k-1}} + \mathbf{B}_{i\lhd_k^j,t_k} \mathbf{U}_{i\lhd_k^j,t_k} \tag{38}$$

Priori covariance prediction:

$$\hat{P}^-_{i\lhd_k^j,t_k} = \mathbf{F}_{i\lhd_k^j,t_k} \hat{P}_{i\lhd_k^j,t_{k-1}} \mathbf{F}^{\mathrm{T}}_{i\lhd_k^j,t_k} + \mathbf{Q}_{i\lhd_k^j,t_k} \tag{39}$$

where the superscript $\blacksquare^-$ is the marker of the prior estimation. In this paper, the measurement and estimated values are denoted by $\blacksquare$ and $\hat{\blacksquare}$, respectively. $\mathbf{F}_{i\lhd_k^j}$ is the state transition model and can be written as:

$$\mathbf{F}_{i\lhd_k^j} = \frac{\partial \mathbf{f}\left(\mathbf{X}_{i\lhd_k^j}, \mathbf{U}_{i\lhd_k^j}\right)}{\partial [\mathbf{X}_{i\lhd_k^j}]^{\mathrm{T}}}$$

$$= \begin{bmatrix} -\tilde{\boldsymbol{\Omega}}_i & \mathbf{0}_{2\times 2} & \boldsymbol{\Gamma}\mathbf{R}^{\beta_i\beta_j}\mathbf{v}_j^{\beta_j} & \mathbf{0}_{2\times 1} & -\mathbf{I}_{2\times 2} & \mathbf{R}^{\beta_i\beta_j} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & -\tilde{\boldsymbol{\Omega}}_i & \mathbf{0}_{2\times 1} & \boldsymbol{\Gamma}\mathbf{R}^{\beta_i\beta_k}\mathbf{v}_k^{\beta_k} & -\mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{R}^{\beta_i\beta_k} \\ \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & 0 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & 0 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & -\tilde{\boldsymbol{\Omega}}_i & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & -\tilde{\boldsymbol{\Omega}}_j & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & -\tilde{\boldsymbol{\Omega}}_k \end{bmatrix} \tag{40}$$

and the control-input model, $\mathbf{B}_{i\lhd_k^j}$, can be obtained by:

$$\mathbf{B}_{i\lhd_k^j} = \begin{bmatrix} -\boldsymbol{\Gamma}\mathbf{P}_{ij}^{\beta_i} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ -\boldsymbol{\Gamma}\mathbf{P}_{ik}^{\beta_i} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ -1 & 1 & 0 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ -1 & 0 & 1 & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} & \mathbf{0}_{1\times 2} \\ -\boldsymbol{\Gamma}\mathbf{v}_i^{\beta_i} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & \mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 1} & -\boldsymbol{\Gamma}\mathbf{v}_i^{\beta_i} & \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 2} & \mathbf{I}_{2\times 2} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 1} & \mathbf{0}_{2\times 1} & -\boldsymbol{\Gamma}\mathbf{v}_i^{\beta_i} & \mathbf{0}_{2\times 2} & \mathbf{0}_{2\times 2} & \mathbf{I}_{2\times 2} \end{bmatrix} \tag{41}$$

The matrix $\mathbf{Q}$ represents the covariance of the process noise. It can be defined based on the covariance matrix of the input noise, as follows [38]:

$$\mathbf{Q} = \mathbf{B}Q\mathbf{B}^{\mathrm{T}} \tag{42}$$

where $Q = diag(\sigma^2_{\omega_i}, \sigma^2_{\omega_j}, \sigma^2_{\omega_k}, \sigma^2_{\mathbf{a}_i}, \sigma^2_{\mathbf{a}_j}, \sigma^2_{\mathbf{a}_k})$ and σ is the standard deviation.

- **Update step**: The predicted state estimate and covariance matrix are updated by comparing the predicted measurement with

the actual measurement. The following equations are used to calculate the updated state estimate and covariance matrix:

$$\mathbf{K}_{t_k} = \mathcal{P}^{-}_{i\triangleleft^j_k,t_k}\mathbf{H}^{\mathrm{T}}_{i\triangleleft^j_k,t_k}\left(\mathbf{H}_{i\triangleleft^j_k,t_k}\mathcal{P}^{-}_{i\triangleleft^j_k,t_k}\mathbf{H}^{\mathrm{T}}_{i\triangleleft^j_k,t_k}+\mathcal{R}_{t_k}\right)^{-1} \tag{43}$$

$$\hat{\mathbf{X}}_{i\triangleleft^j_k,t_k} = \hat{\mathbf{X}}^{-}_{i\triangleleft^j_k,t_k} + \mathbf{K}_{t_k}\left(\tilde{\mathbf{y}}_{i\triangleleft^j_k} - \mathbf{h}\left(\hat{\mathbf{X}}_{i\triangleleft^j_k}\right)\right) \tag{44}$$

$$\mathcal{P}_{i\triangleleft^j_k,t_k} = \left(\mathbf{I}-\mathbf{K}_{t_k}\mathbf{H}_{i\triangleleft^j_k,t_k}\right)\mathcal{P}^{-}_{i\triangleleft^j_k,t_k} \tag{45}$$

where $\tilde{\mathbf{y}}$ is the measurement vector and $\mathcal{R} = diag(\sigma^2_{d_{ij}}, \sigma^2_{d_{ik}}, \sigma^2_{\mathbf{R}_{jk}}, \sigma^2_{\mathbf{v}_i}, \sigma^2_{\mathbf{v}_j}, \sigma^2_{\mathbf{v}_k})$ is its covariance matrix. Also, $\mathbf{H}_{i\triangleleft^j_k}$ is the Jacobian matrix of the measurement model that can be obtained by:

$$\mathbf{H}_{i\triangleleft^j_k} = \frac{\partial \mathbf{h}(\mathbf{X}_{i\triangleleft^j_k})}{\partial[\mathbf{X}_{i\triangleleft^j_k}]^{\mathrm{T}}} = \begin{bmatrix} \frac{[\mathbf{P}^{\beta_i}_{ij}]^{\mathrm{T}}}{\|\mathbf{P}^{\beta_i}_{ij}\|} & \mathbf{0}_{1\times2} & 0 & 0 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{1\times2} & \frac{[\mathbf{P}^{\beta_i}_{ik}]^{\mathrm{T}}}{\|\mathbf{P}^{\beta_i}_{ik}\|} & 0 & 0 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ -\mathbf{I}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 1 & 0 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & 0 & 1 & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times2} & \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times1} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{I}_{2\times2} \end{bmatrix} \tag{46}$$

4.2. The configuration of cooperation among UAVs

All three UAVs in a group perform their estimation sequentially (in three steps). According to Fig. 2, the values of the measurement vector, $\tilde{\mathbf{y}}$, for all three UAVs are obtained with the help of measurements and estimation of others. The measured velocity, heading angle and distances between every quadrotor are denoted by $\tilde{\mathbf{v}}, \tilde{\psi}$, and $\tilde{d}$, respectively. Following are details about determining the values of the $\tilde{\mathbf{y}}$ in each step.

- **Step 1** $\tilde{\mathbf{y}}_{i\triangleleft^j_k}$ can be written as follows:

$$\tilde{\mathbf{y}}_{i\triangleleft^j_k} = \left[\tilde{d}_{ij}, \tilde{d}_{ik}, \mathbf{P}^{\beta_i}_{ik} - \mathbf{P}^{\beta_i}_{ij}, \Delta\tilde{\psi}_{ij}, \Delta\tilde{\psi}_{ik}, \tilde{\mathbf{v}}^{\beta_i}_i, \tilde{\mathbf{v}}^{\beta_j}_j, \tilde{\mathbf{v}}^{\beta_k}_k\right]^{\mathrm{T}} \tag{47}$$

where $\mathbf{P}^{\beta_i}_{ik} - \mathbf{P}^{\beta_i}_{ij} = \mathbf{P}^{\beta_i}_{jk}$ can be estimated as follows:

$$\mathbf{P}^{\beta_i}_{jk} = \tilde{d}_{jk}\frac{[\hat{\mathbf{R}}^{\beta_j\beta_i}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_j}_{jk} - [\hat{\mathbf{R}}^{\beta_k\beta_i}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_k}_{kj}}{\left\|[\hat{\mathbf{R}}^{\beta_j\beta_i}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_j}_{jk} - [\hat{\mathbf{R}}^{\beta_k\beta_i}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_k}_{kj}\right\|} \tag{48}$$

where $\hat{\mathbf{R}} = \mathbf{R}(\Delta\hat{\psi})$ represents the estimated rotation matrix. $\hat{\mathbf{u}}$ is the unit vector of estimated $\mathbf{P}$, denoted by $\hat{\mathbf{P}}$. As $\Delta\hat{\psi}$ and $\hat{\mathbf{P}}$ are calculated in the previous sample time, it is better to use the compensation values ($\hat{\psi}^-$ and $\hat{\mathbf{P}}^-$) of them. The compensation process can be performed by propagating the following dynamic equations.

$$\begin{bmatrix}\dot{\mathbf{P}}^{\beta_j}_{jk}\\ \Delta\dot{\psi}_{ji}\end{bmatrix} = \begin{bmatrix}-\mathbf{v}^{\beta_j}_j + \mathbf{R}^{\beta_j\beta_k}\mathbf{v}^{\beta_k}_k - \tilde{\boldsymbol{\Omega}}_j\mathbf{P}^{\beta_j}_{jk}\\ \tilde{\omega}_i - \tilde{\omega}_j\end{bmatrix} \tag{49}$$

$$\begin{bmatrix}\dot{\mathbf{P}}^{\beta_k}_{kj}\\ \Delta\dot{\psi}_{ki}\end{bmatrix} = \begin{bmatrix}-\mathbf{v}^{\beta_k}_k + \mathbf{R}^{\beta_k\beta_j}\mathbf{v}^{\beta_j}_j - \tilde{\boldsymbol{\Omega}}_k\mathbf{P}^{\beta_k}_{kj}\\ \tilde{\omega}_i - \tilde{\omega}_k\end{bmatrix} \tag{50}$$

- **Step 2** $\tilde{\mathbf{y}}_{j\triangleleft^i_k}$ can be written as follows:

$$\tilde{\mathbf{y}}_{j\triangleleft^i_k} = \left[\tilde{d}_{ji}, \tilde{d}_{jk}, \mathbf{P}^{\beta_j}_{jk} - \mathbf{P}^{\beta_j}_{ji}, \Delta\tilde{\psi}_{ji}, \Delta\tilde{\psi}_{jk}, \tilde{\mathbf{v}}^{\beta_j}_j, \tilde{\mathbf{v}}^{\beta_i}_i, \tilde{\mathbf{v}}^{\beta_k}_k\right]^{\mathrm{T}} \tag{51}$$

where $\mathbf{P}^{\beta_j}_{jk} - \mathbf{P}^{\beta_j}_{ji} = \mathbf{P}^{\beta_j}_{ik}$ can be estimated as follows:

$$\mathbf{P}^{\beta_j}_{ik} = \tilde{d}_{ik}[\hat{\mathbf{R}}^{\beta_i\beta_j}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_i}_{ik} \tag{52}$$

- **Step 3** $\tilde{\mathbf{y}}_{k\triangleleft^i_j}$ can be written as follows:

$$\tilde{\mathbf{y}}_{k\triangleleft^i_j} = \left[\tilde{d}_{ki}, \tilde{d}_{kj}, \mathbf{P}^{\beta_k}_{kj} - \mathbf{P}^{\beta_k}_{ki}, \Delta\tilde{\psi}_{ki}, \Delta\tilde{\psi}_{kj}, \tilde{\mathbf{v}}^{\beta_k}_k, \tilde{\mathbf{v}}^{\beta_i}_i, \tilde{\mathbf{v}}^{\beta_j}_j\right]^{\mathrm{T}} \tag{53}$$

where $\mathbf{P}^{\beta_k}_{kj} - \mathbf{P}^{\beta_k}_{ki} = \mathbf{P}^{\beta_k}_{ij}$ can be estimated as follows:

$$\mathbf{P}^{\beta_k}_{ij} = \tilde{d}_{ij}\frac{[\hat{\mathbf{R}}^{\beta_i\beta_k}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_i}_{ij} - [\hat{\mathbf{R}}^{\beta_j\beta_k}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_j}_{ji}}{\left\|[\hat{\mathbf{R}}^{\beta_i\beta_k}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_i}_{ij} - [\hat{\mathbf{R}}^{\beta_j\beta_k}]^{\mathrm{T}}\hat{\mathbf{u}}^{\beta_j}_{ji}\right\|} \tag{54}$$

5. Verification through simulations

Several simulation tests are conducted to examine the performance of the proposed estimation scheme, including noise-free studies on observable/unobservable situations, disturbances, and disconnection studies in the presence of noisy measurements. While several modern control algorithms are available for quadrotors, such as Model Predictive Control (MPC) [39] and sliding mode control, this paper utilizes a simpler method, a Proportional–Derivative (PD) controller [40], to control the position of the quadrotor by controlling its attitude. The dynamics of the quadrotors in six degrees of freedom are simulated using the model and control method presented in [40]. A high-level representation of the system model is shown in Fig. 3. The quadrotor has a total mass of 1.25 kg, with inertia moments $\mathrm{I}_{xx} = 0.0232, \mathrm{I}_{yy} = 0.0232$, and $\mathrm{I}_{zz} = 0.0468$. Each arm of the quadrotor has a length of 26 cm. The PD controller parameters are set as follows: for position control, the proportional gain and derivative gain of each channel of control are set to 0.8 and 0.5, respectively. For roll and pitch control, the proportional gain and derivative gain are set to 750 and 45, respectively. For yaw control, the proportional gain and derivative gain are set to 70 and 20. The simulations are run at a frequency of 50 Hz. To simulate quadrotors movement on a horizon plane, the height changes are kept small. In these simulations, the three quadrotors move on the analytically deterministic trajectory.

5.1. Unobservable situations

In this section, the simulation scenarios are designed to confirm the observability analysis conclusions. The critical scenarios of moving in parallel and being stationary are investigated in these scenarios. These simulations will determine whether the EKFs are convergent or not. All the quadrotors know the measurements without noise. The yaw rates and headings of every quadrotors are kept at 0 rad/s and 0 rad, respectively. In EKFs, 0.1 of the real value of relative positions is considered as the initial guess. The initial estimation errors for heading angles are $\Delta\hat{\psi}_{ij} = 30°, \Delta\hat{\psi}_{ik} = -20°$, and $\Delta\hat{\psi}_{jk} = -50°$. Since the velocities of quadrotors are not of interest in the analysis, they are initialized to their actual values. For each scenario, the following trajectory references in the reference inertial frame are defined.

- **Stationary:** In this scenario, the quadrotor are hovering on points that are collinear and non-collinear.
  - Collinear

$$\mathbf{t}^i_1 = \begin{bmatrix}0 & 0\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^j_1 = \begin{bmatrix}3 & 0\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^k_1 = \begin{bmatrix}6 & 0\end{bmatrix}^{\mathrm{T}} \tag{55}$$

  - Non-collinear

$$\mathbf{t}^i_2 = \begin{bmatrix}0 & 1\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^j_2 = \begin{bmatrix}2 & 0\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^k_2 = \begin{bmatrix}3 & -1\end{bmatrix}^{\mathrm{T}} \tag{56}$$

- **Parallel path:** In this scenario, the three quadrotors fly parallel with the presence or absence of collinearity.
  - Collinear

$$\mathbf{t}^i_3 = \begin{bmatrix}t & 3\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^j_3 = \begin{bmatrix}t & 0\end{bmatrix}^{\mathrm{T}},\ \mathbf{t}^k_3 = \begin{bmatrix}t & -3\end{bmatrix}^{\mathrm{T}} \tag{57}$$

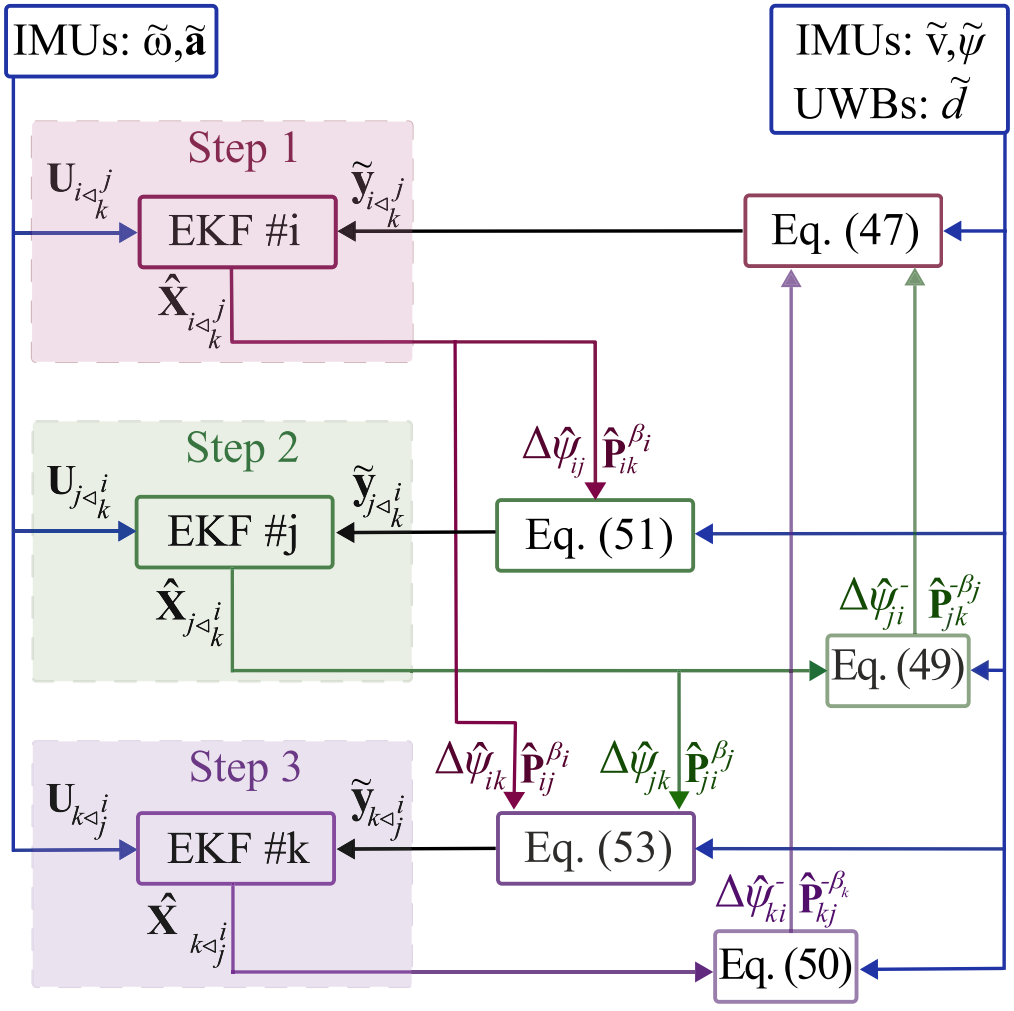


Fig. 2. Framework of the proposed estimation strategy.

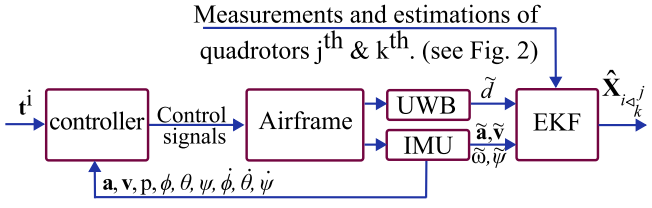


Fig. 3. System model for ith quadrotor.

– Non-collinear

$$\mathbf{t}_4^i = \begin{bmatrix} t & 3 \end{bmatrix}^{\mathrm{T}},\ \mathbf{t}_4^j = \begin{bmatrix} 2+t & 0 \end{bmatrix}^{\mathrm{T}},\ \mathbf{t}_4^k = \begin{bmatrix} 1+t & -3 \end{bmatrix}^{\mathrm{T}} \tag{58}$$

where $\mathbf{t}_n^i$ represents the nth trajectory of the ith quadrotor and t is the time.

Figs. 4 and 5 show the estimation of relative position and heading angle for a stationary and collinear trajectory ($\mathbf{t}_1$), respectively. However, the results for a stationary and non-collinear trajectory ($\mathbf{t}_2$) are shown in Fig. 6. Similarly, the results of the parallel path while the quadrotors are collinear($\mathbf{t}_3$) are presented in Figs. 7 and 8, as well as while they are non-collinear ($\mathbf{t}_4$) are illustrated in Fig. 9.

In the collinear situations, $\mathbf{t}_1$ and $\mathbf{t}_3$, none of the three sets of observability conditions are met, while in the non-collinear situations, $\mathbf{t}_2$ and $\mathbf{t}_4$, the third set of observability conditions, Eq. (33), are satisfied. Therefore, the estimation results demonstrate that, as depicted in

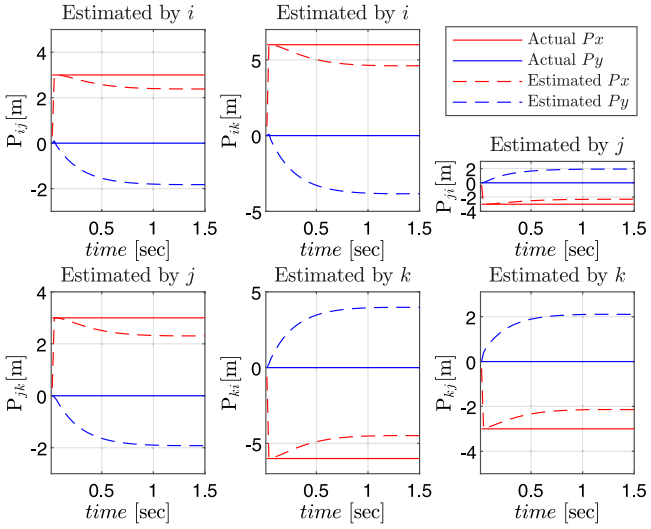


Fig. 4. Estimation of relative position for a stationary and collinear trajectory ($\mathbf{t}_1$).

Figs. 4 and 7, the values have converged to their actual values, whereas in the estimation results shown in Figs. 6 and 9, this convergence has not occurred. Moreover, based on the results obtained from the observability analysis Section 3.2, it is concluded that the estimation

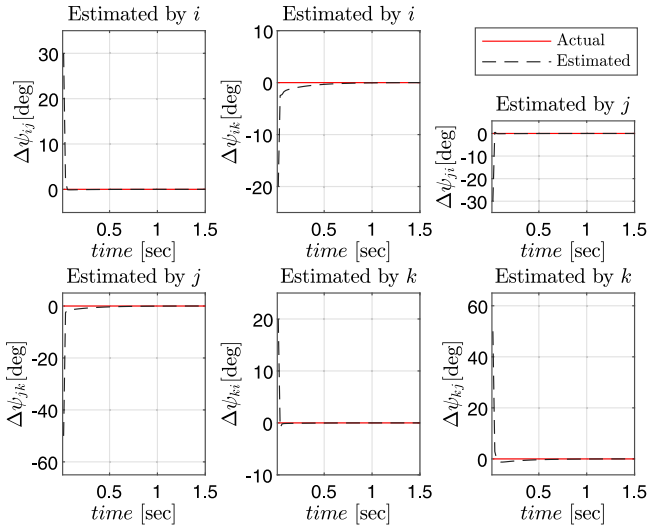


Fig. 5. Estimation of relative heading angle for a stationary and collinear trajectory ($\mathbf{t}_1$).

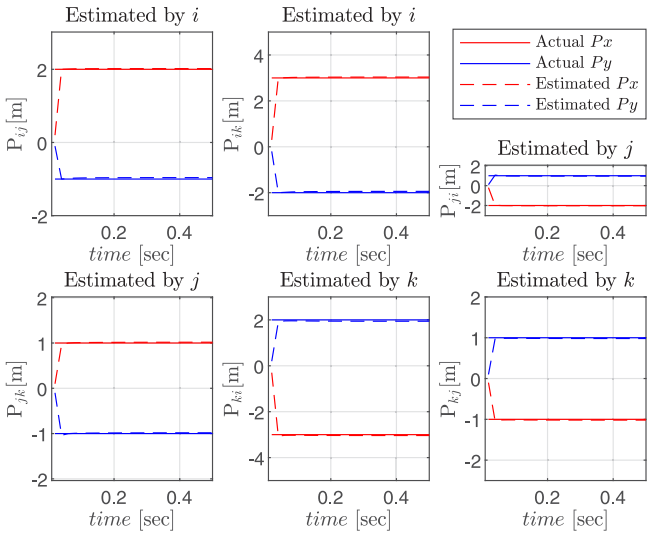


Fig. 6. Estimation of relative position for a stationary and non-collinear trajectory ($\mathbf{t}_2$).

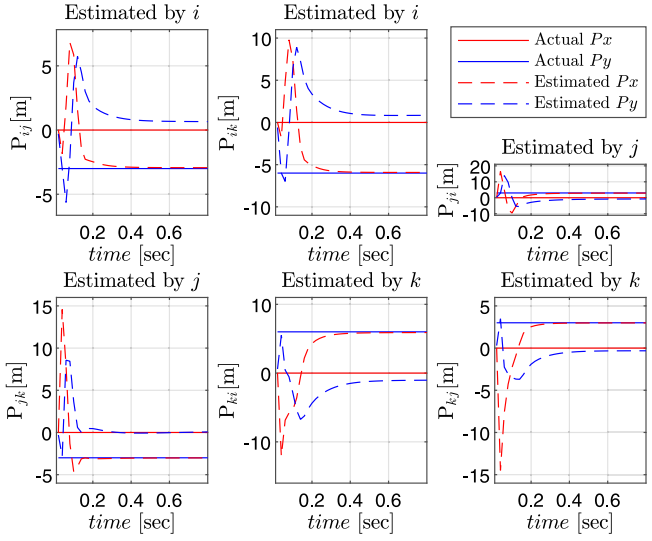


Fig. 7. Estimation of relative position for a parallel and collinear trajectory ($\mathbf{t}_3$).

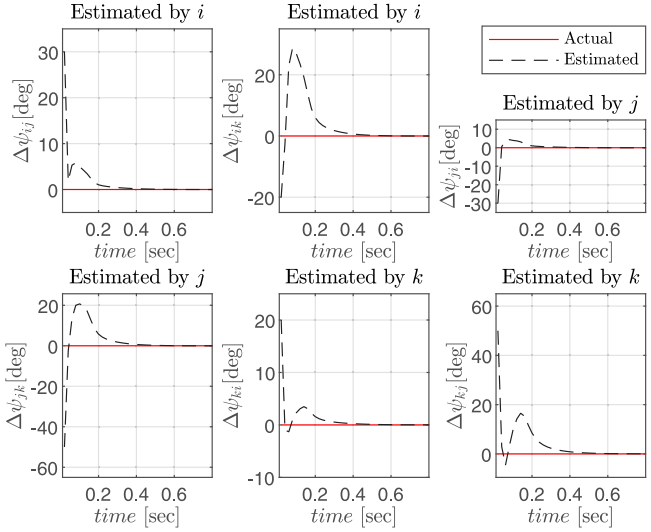


Fig. 8. Estimation of relative heading angle for a parallel and collinear trajectory ($\mathbf{t}_3$).

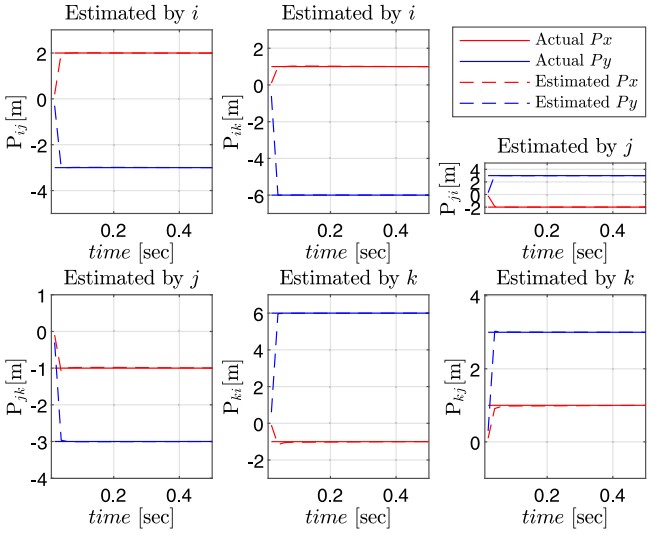


Fig. 9. Estimation of relative position for a parallel and non-collinear trajectory ($\mathbf{t}_4$).

of the relative heading angle remains observable under all conditions. The simulation results support this finding as well. As can be seen, the relative heading angles are converged to their actual values in all scenarios.

5.2. Noise-free and noisy measurements study

This section investigates the performance of the proposed method in the presence of noise. In order to compare the performance of our method with other approaches, we also generated results for this scenario using a Decentralized estimation strategy based on Consensus EKF (DCEKF), as proposed in [32]. In terms of the initialization of EKFs, the initial value of the relative positions is set to 0.1 of its true value, while the other states of the model are initialized to their true values. Fig. 10 illustrates the circular trajectories the quadrotors follow (for 22 s), as defined below.

$$\begin{cases} \mathbf{t}_5^i = 2\begin{bmatrix}\cos(0.3t) & \sin(0.3t)\end{bmatrix}^{\mathrm{T}} \\ \mathbf{t}_5^j = 3\begin{bmatrix}\cos(0.3t+\pi/4) & \sin(0.3t+\pi/4)\end{bmatrix}^{\mathrm{T}} \\ \mathbf{t}_5^k = 4\begin{bmatrix}\cos(0.3t+\pi/2) & \sin(0.3t+\pi/2)\end{bmatrix}^{\mathrm{T}} \end{cases} \tag{59}$$

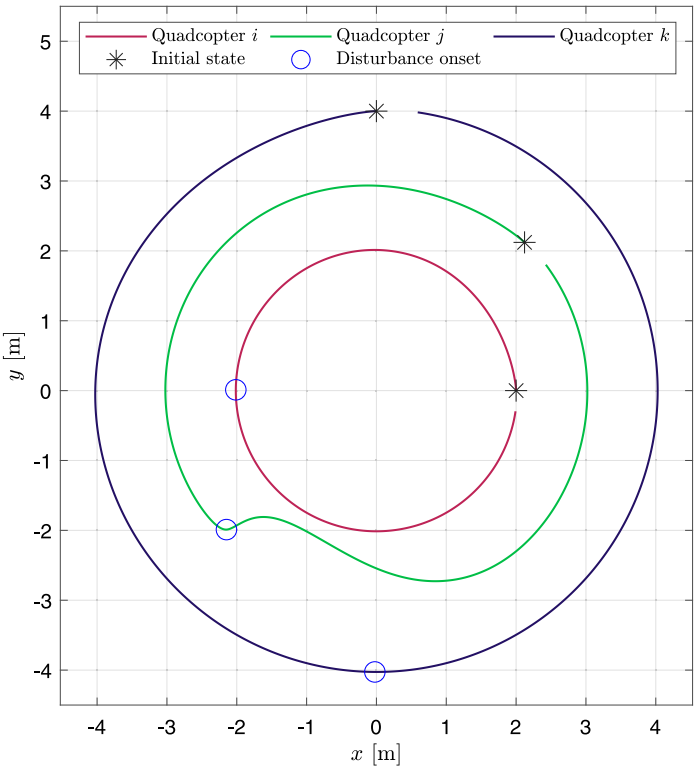


Fig. 10. The paths followed by the quadrotors in trajectory $\mathbf{t}_5$. The acceleration disturbance is applied to quadrotor j at the 12th second of the simulation.

Figs. 11 and 12 present the simulation results for a duration of 100 to 170 s. During the initial 100 s of the simulation, both methods demonstrate accurate estimation. However, as the simulation progresses, the differences (due to accumulated error of DCEKF method) between the two methods become evident. Additionally, noisy measurements are modeled as $\tilde{\blacksquare} = \blacksquare_{tv} + N(0, \sigma)$, where $\blacksquare_{tv}$ denotes true value and N represents the normal white noise with a zero mean and standard deviation σ. The standard deviation of the UWB measurement noise is considered to be $\sigma_d = 0.1$ m. Similarly, the standard deviation of IMU measurement noise is considered as $\sigma_{\mathbf{a}} = 0.02\mathbf{I}_{2\times 2}$ m/sec^2 and $\sigma_\omega = 0.03$ rad/sec. Figs. 13 and 14 illustrate the performance of both methods with noisy measurements. Our method outperforms DCEKF due to its ability to resist arising errors. Furthermore, using an appropriate cooperative configuration (presented in Section 4.2) which incorporates the sharing and fusion of information from different sources, enables more accurate estimation results with less noise compared to the DCEKF method. It should also be acknowledged that as the noise amplitude increases, the quality of the results can be deteriorated [38].

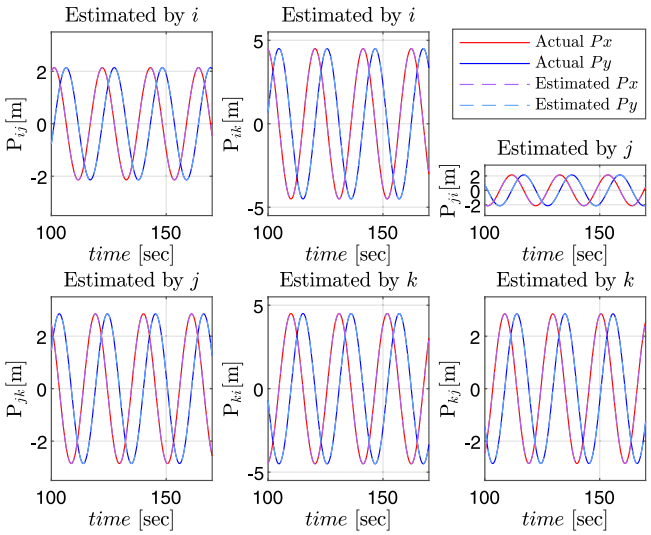


Fig. 11. Estimation of the relative position for free-noise measurements study.

5.3. Disturbance study

This section investigates the performance of the proposed method under the influence of applied disturbances. The acceleration disturbance $\mathbf{a}'_j = [0,\ 4e^{-(0.1|12-t|)}]^{\mathrm{T}}$ is applied to the jth quadrotor. The results of the relative position estimation are presented in Fig. 15. The results demonstrate that our method outperforms the DCEKF approach (shown in Fig. 16) in terms of convergence to the actual values, even in the presence of disturbances. This is because our method incorporates geometrical constraints into the relative motion model, allowing each EKF in each quadrotor to solve the estimation problem subject to these constraints. In contrast, in the DCEKF approach, the EKFs solve the estimation problem without considering geometrical constraints, and these constraints are only considered in the consensus procedure used to justify the estimation results and improve their accuracy.

5.4. Disconnection study

In this section, the proposed method is evaluated within a scenario where the quadrotors experience disconnection. The investigation analyzes the behavior of the estimation process when communication between quadrotors is lost, and evaluates the impact of such disconnections on the accuracy of estimation results. During this scenario, the quadrotors follow the $\mathbf{t}_5$ trajectory while experiencing a loss of communication between 5 and 8 s. To address this issue, the last information received from each quadrotor prior to the disconnection is held and used in the estimation process until communication is restored. The results demonstrate that our method (Fig. 17) outperforms

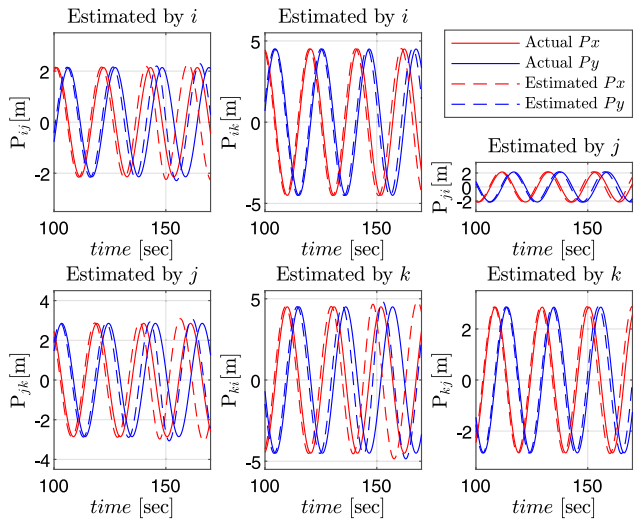


Fig. 12. Generated estimation results of the DCEKF method [32] for free-noise measurements study.

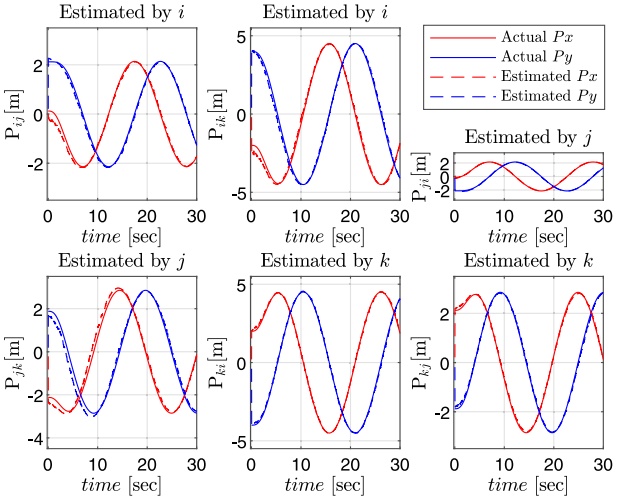


Fig. 13. Estimation of the relative position for noisy measurements study.

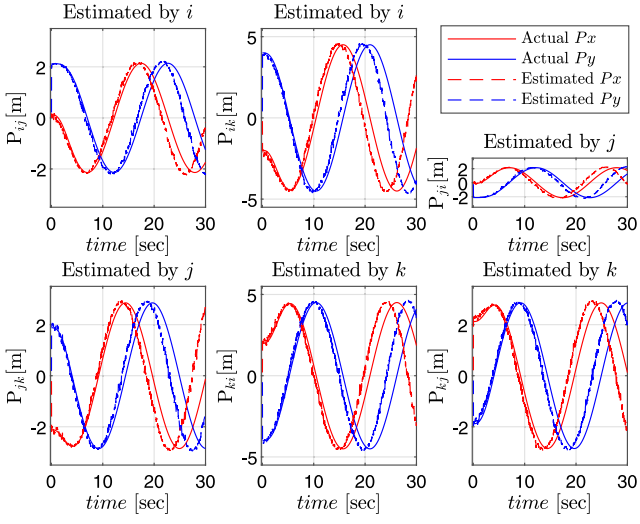


Fig. 14. Generated estimation results of the DCEKF method [32] for noisy measurements study.

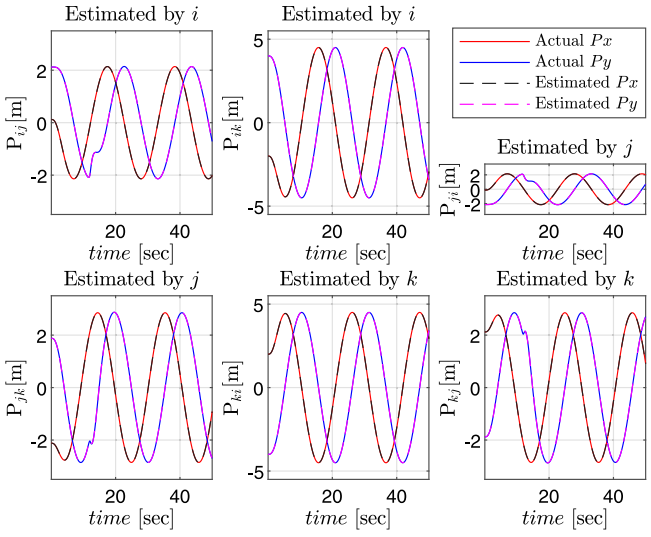


Fig. 15. Estimation of the relative position for disturbance study.

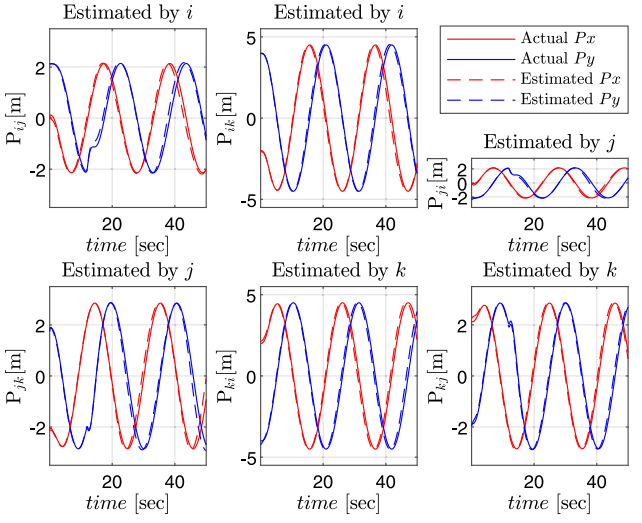


Fig. 16. Generated estimation results of the DCEKF method [32] for disturbance study.

the DCEKF approach (Fig. 18) even when the communication between quadrotors is lost. Due to direct incorporation of geometry constraints during the estimation process, the proposed method enables effective compensation for disconnection errors and convergence to true values. The DCEKF approach applies geometry constraints as a correction term after estimation, which limits its ability to fully compensate for disconnection errors.

6. Conclusion

This paper presented a novel algorithm for cooperative relative localization in groups of quadrotors consisting of three members, based on relative range and inertial measurements. The proposed method employs distributed extended Kalman filters on each quadrotor to solve the estimation problem. To facilitate the use of EKF subject to the geometrical topology of the quadrotor system, a relative motion model is proposed. The observability analysis shows that the proposed scheme is able to estimate even in stationary and parallel scenarios if the quadrotors are not collinear. The cooperative approach, which incorporates the sharing and fusion of information from different sources, enables more accurate and reliable estimation results with less noise.

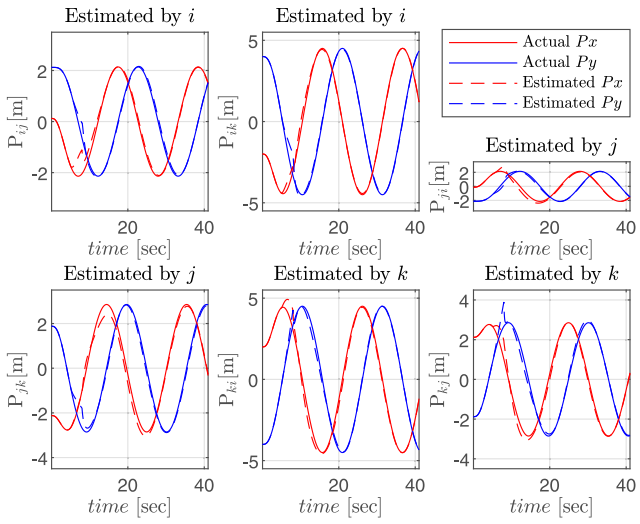


Fig. 17. Estimation of the relative position for disconnection study.

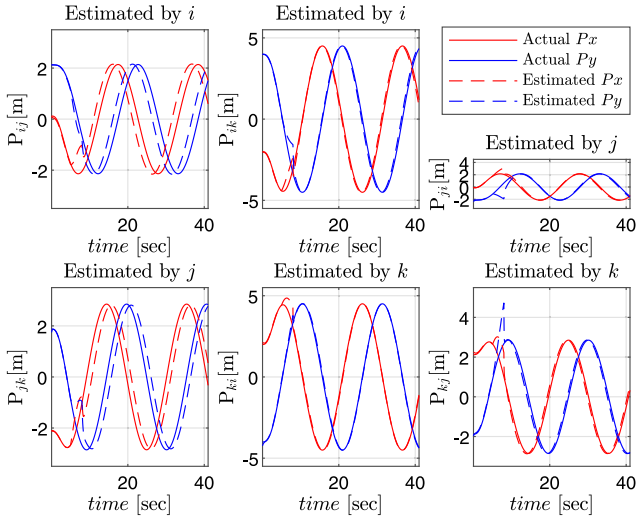


Fig. 18. Generated estimation results of the DCEKF method [32] for disconnection study.

The simulation results demonstrate the effectiveness of our approach in reducing the impact of noise, disturbance, and disconnection in the different scenarios.

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Ethics statement

This research was conducted with adherence to ethical guidelines and protocols.

CRediT authorship contribution statement

Meysam Alizad: Writing original draft, Validation, Methodology, Investigation, Conceptualization. **Hadi Nobahari:** Writing review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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